

Workshop 05.2: Electromagnetic- Thermal Coupling



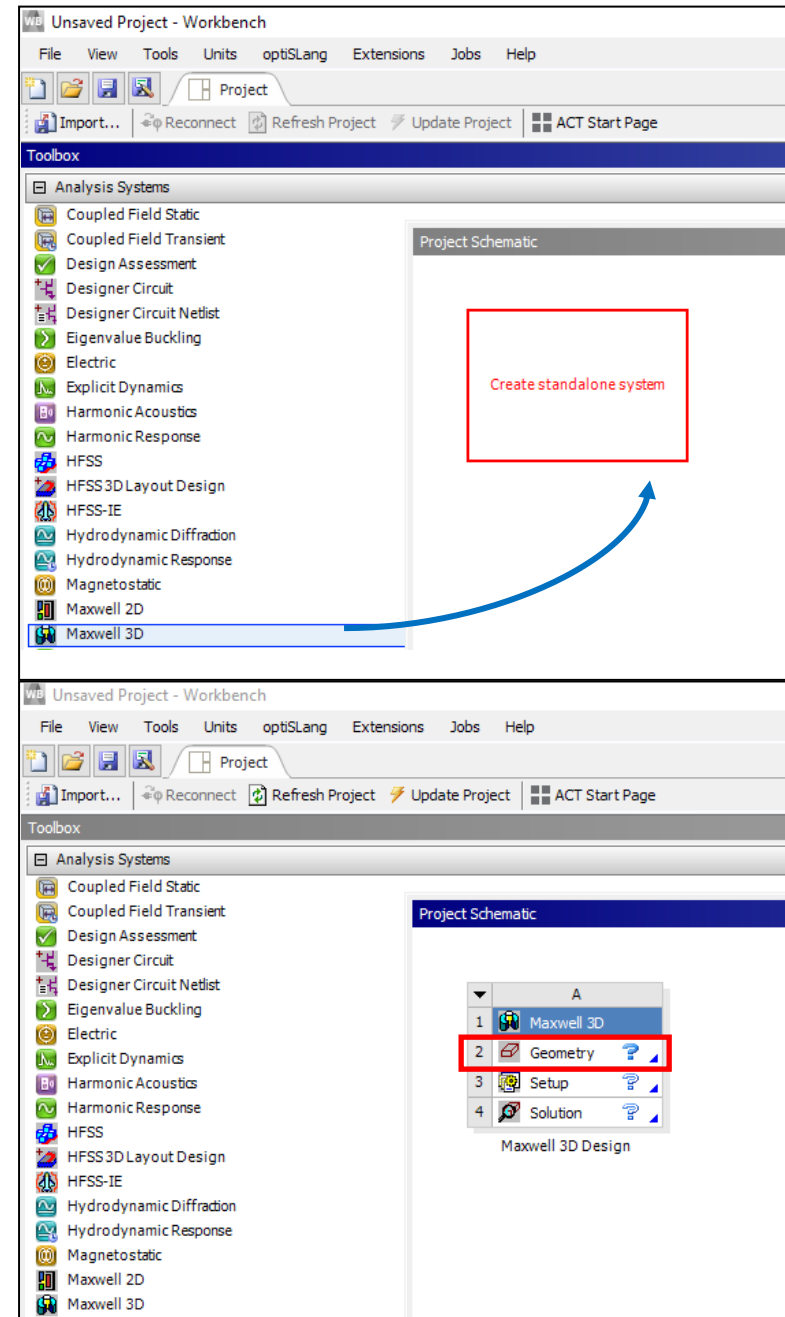
Overview

- **Electromagnetic – Thermal Coupling**

- This workshop will discuss how to use the Ansys Maxwell inside Ansys Workbench to study temperature distribution due to ohmic losses
- The following tasks will be performed:
 - The Geometry of a simple Wave Coil will be imported directly in Maxwell
 - Alternatively, the same Geometry will be created through the 3D Maxwell Modeler
 - An Electromagnetic Magnetostatic analysis will be carried out
 - The Maxwell Solutions will be sent to a Thermal Steady State analysis
 - The Thermal analysis will be realized
 - The temperature distribution will be shown

Project Startup

- Create the Project
 - Start Workbench
 - *Start → All Programs → ANSYS 2020 R1 → Workbench 2020 R1*
 - Expand Analysis Systems and drag and drop a Maxwell 3D system into the Project Schematic
 - Double click on the Maxwell 3D Design Geometry cell (**A2**) to start Ansys Electronics Desktop Interface

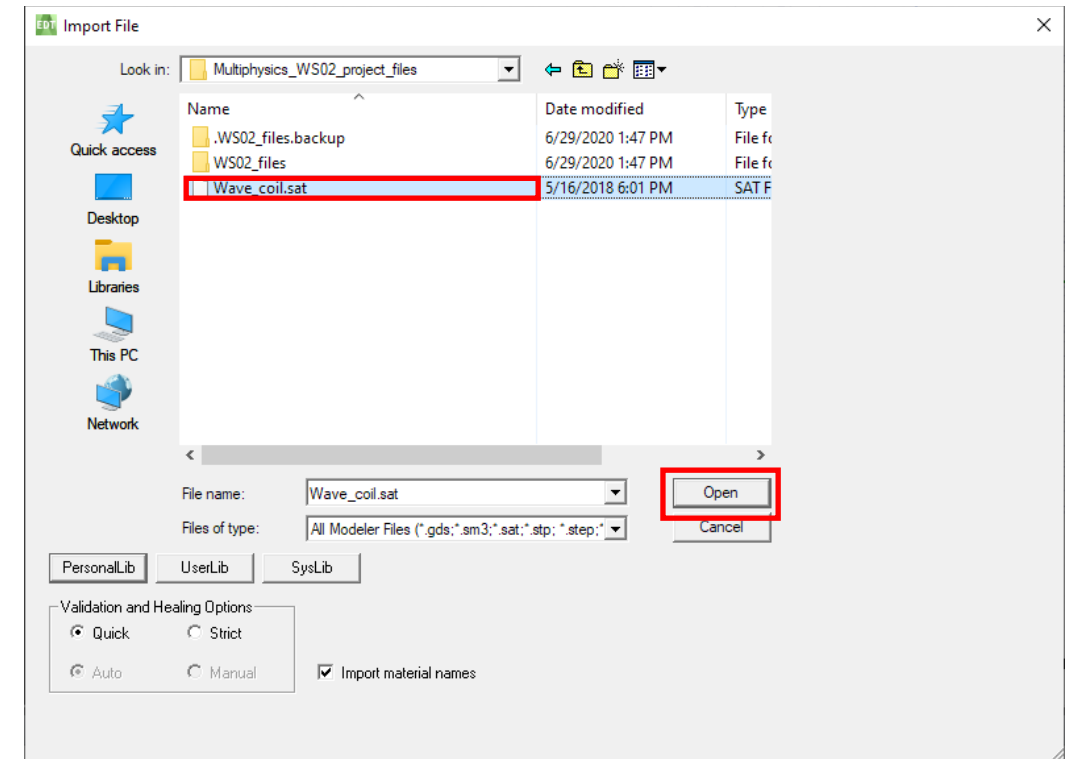
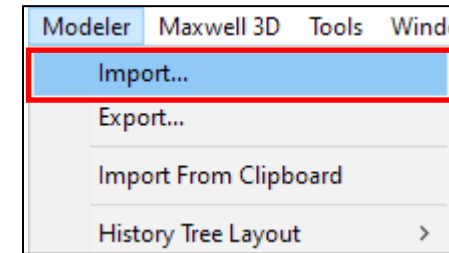


Import

- Import a native ACIS .sat file of Coil Model
 - Choose menu item **Modeler** → **Import**
 - Browse and locate the file “Wave_coil.sat”
 - Select it and press Open to import it

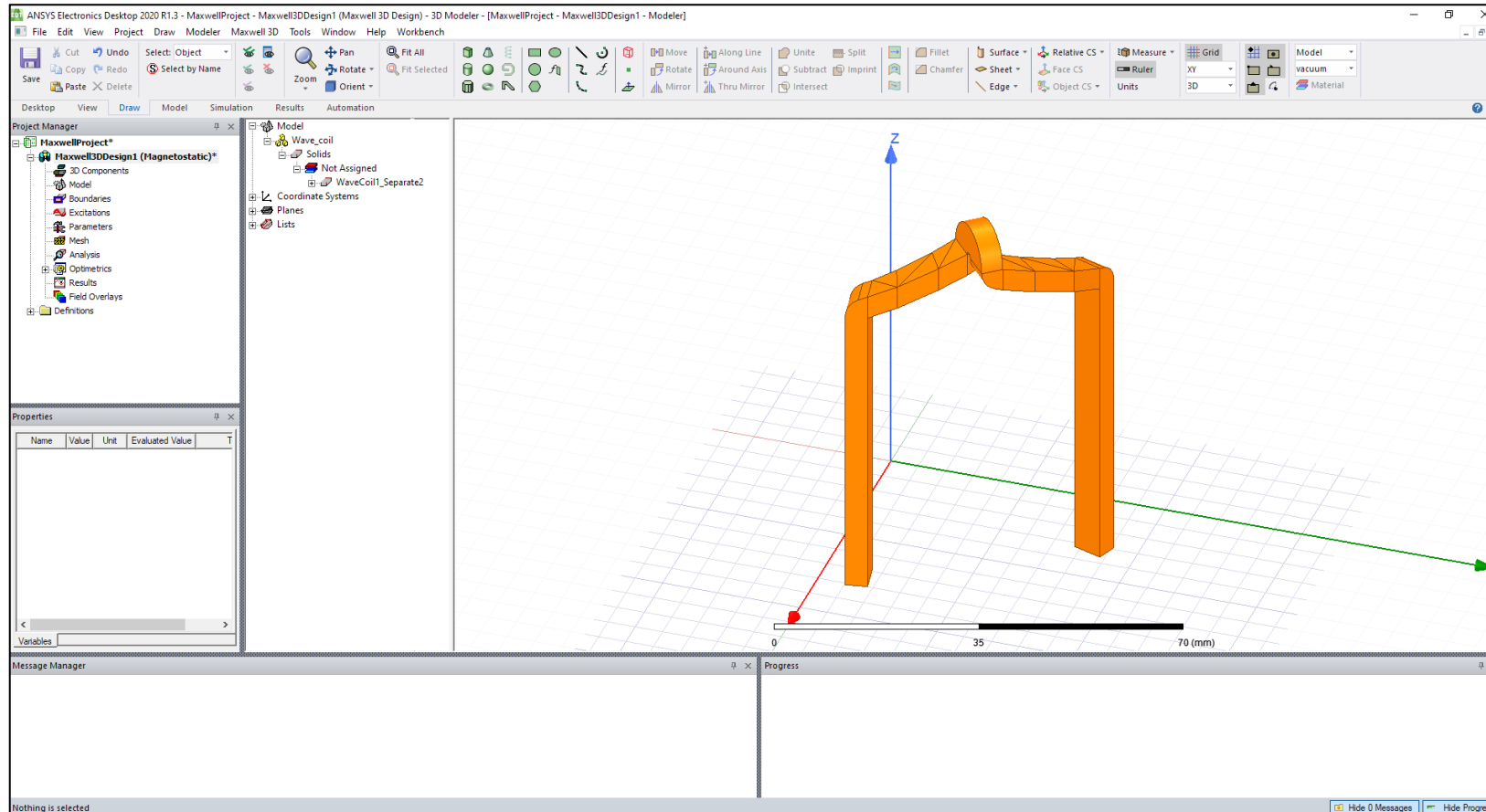
Note 1: the ACIS format is the native Maxwell format and does not require any additional license to be imported

Note 2: in case you would like to build the geometry directly within Maxwell, please go to slide 6



Imported Geometry

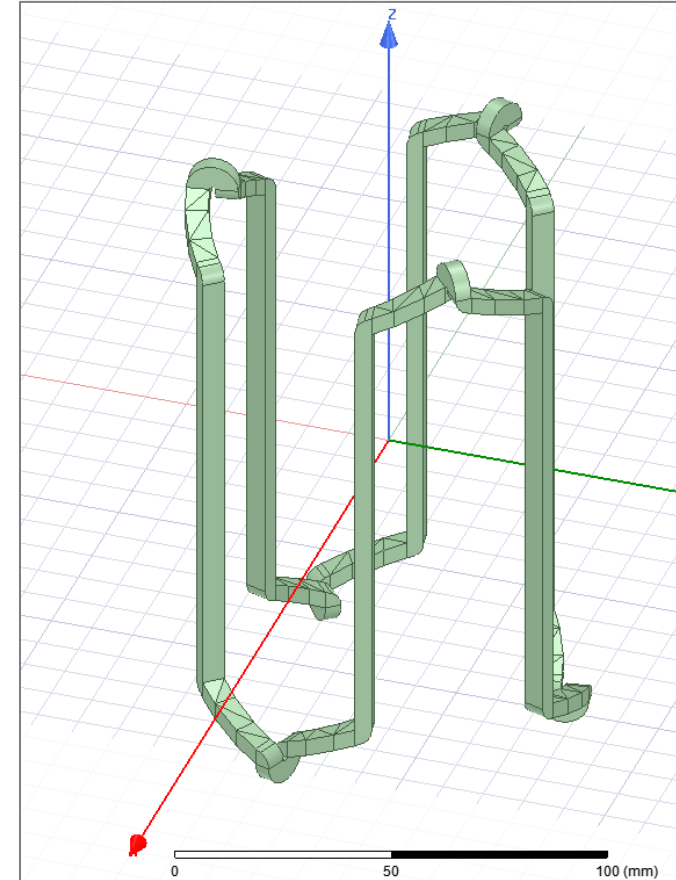
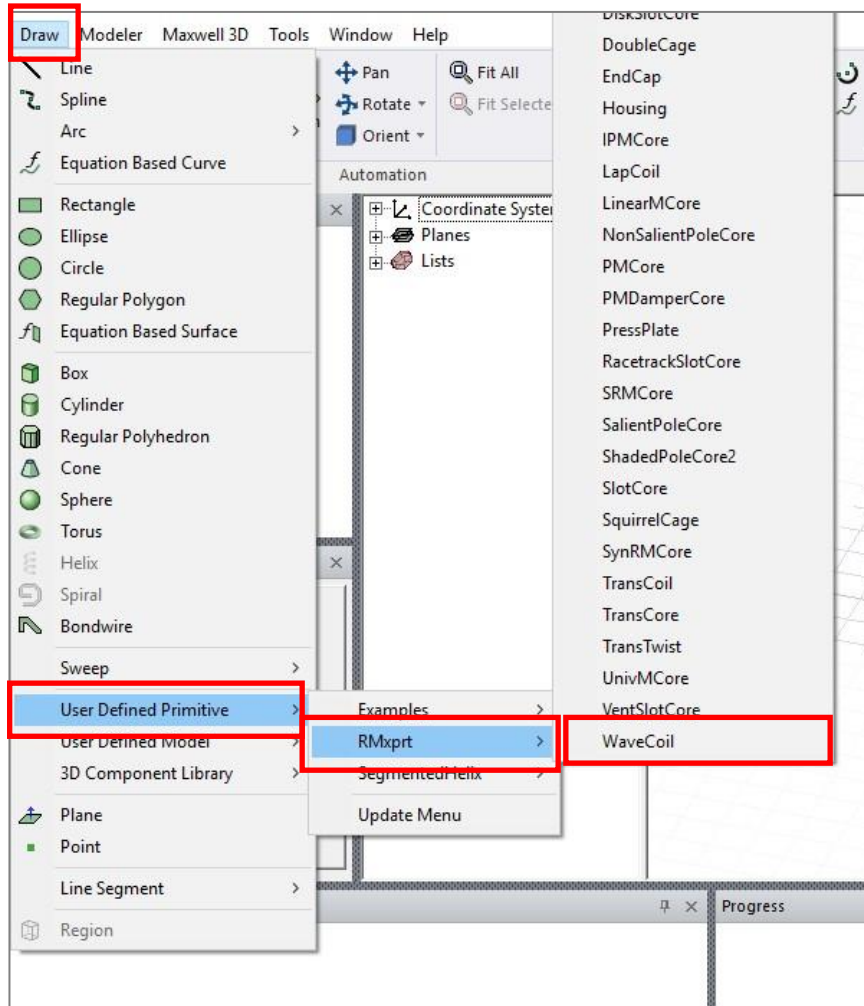
- The Geometry is automatically imported and created



Note: if you are not interested in the Alternative Procedure, skip next slides and go to slide 11

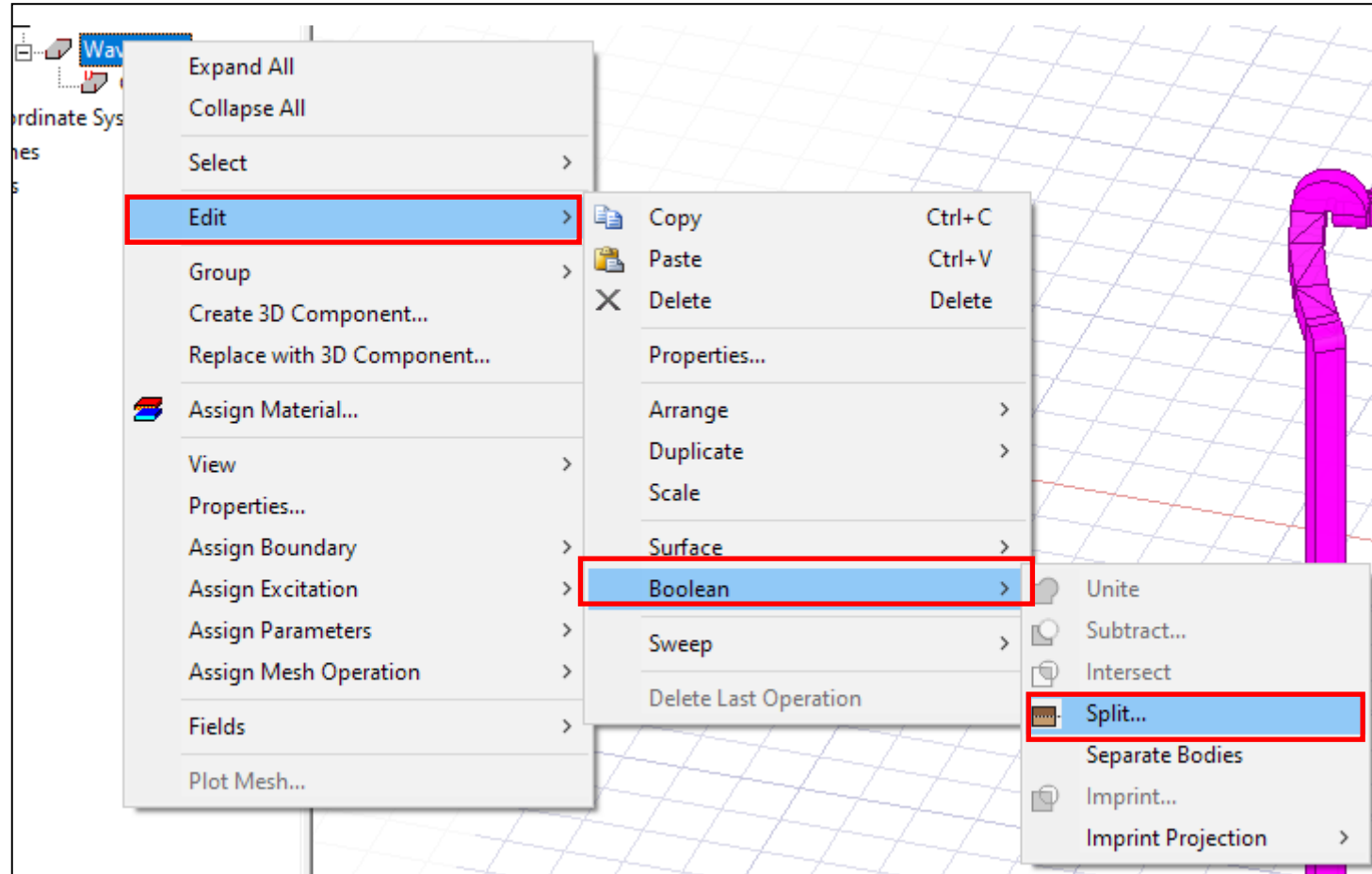
Alternative Procedure

- Choose the menu item **Draw** → **User Defined Primitive** → **RMxpprt** → **WaveCoil**
- Press **OK** on the dialogue window



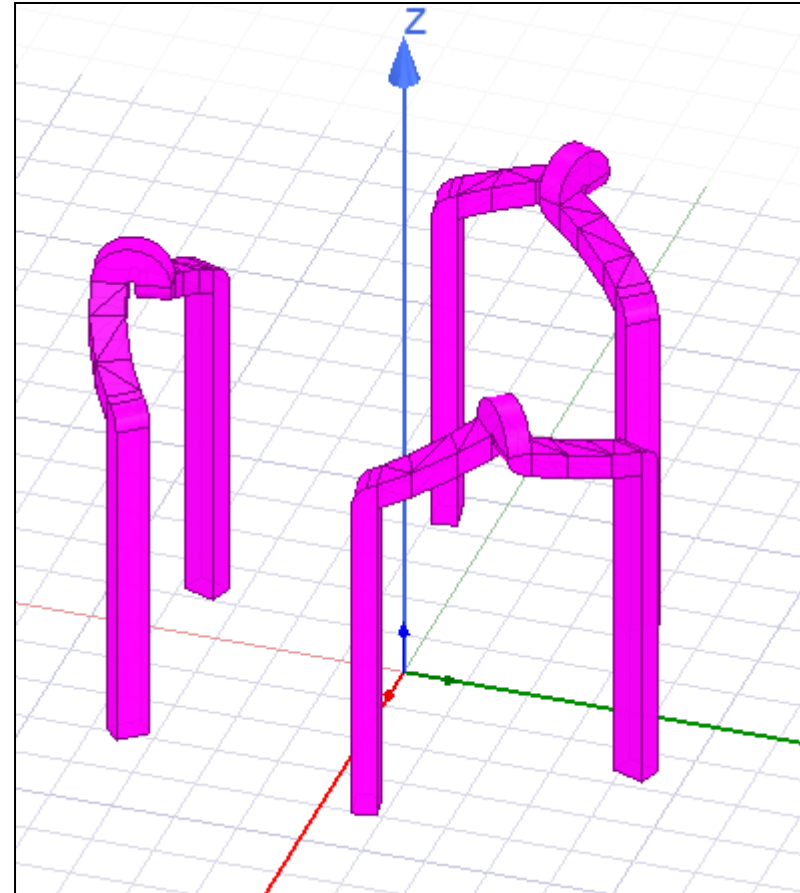
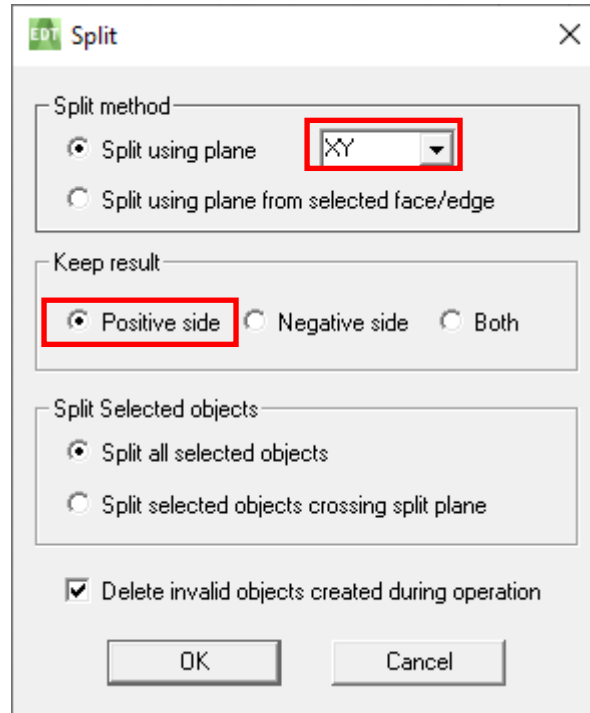
Alternative Procedure

- RMB on the just created *WaveCoil1* → *Edit* → *Boolean* → *Split*



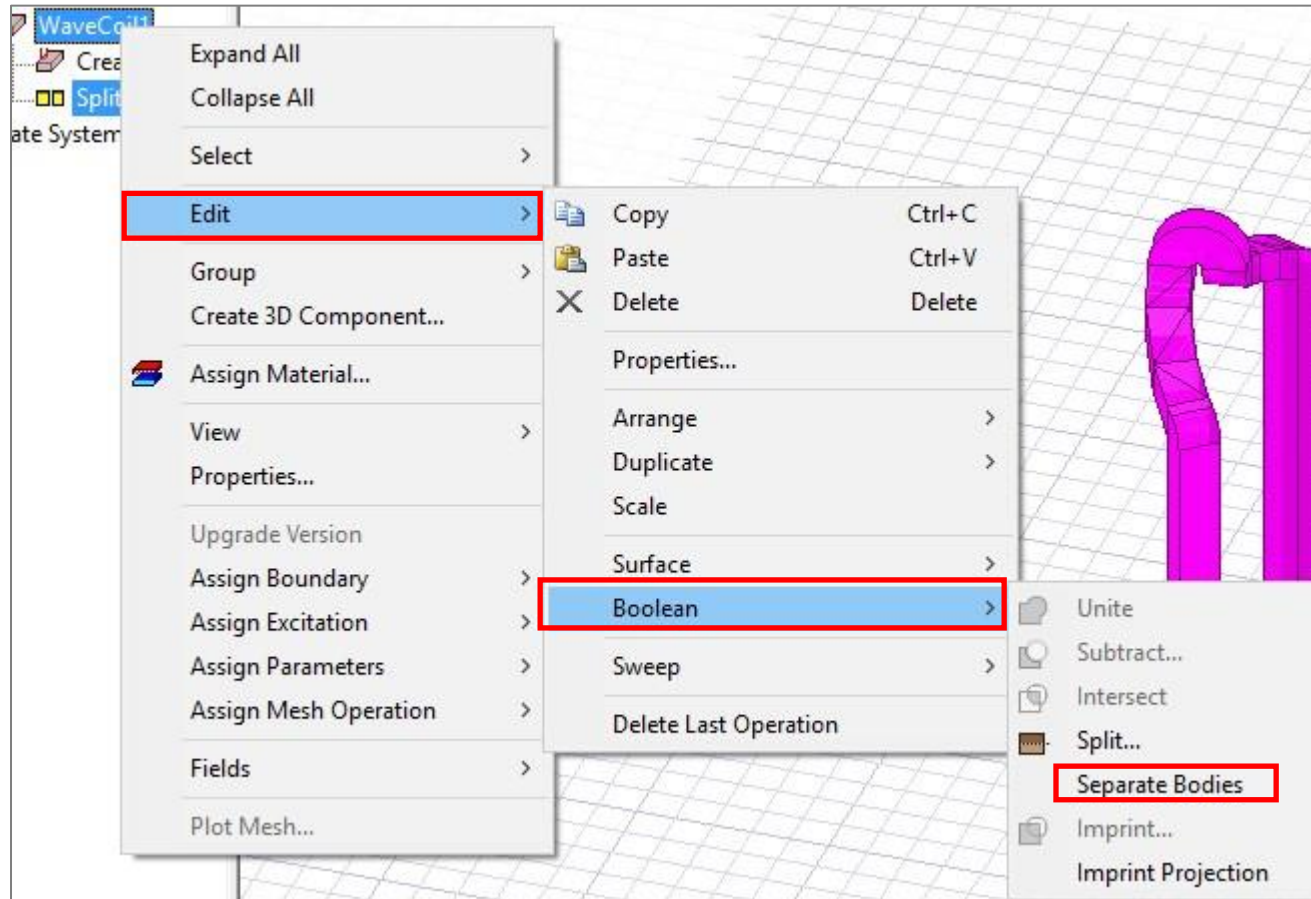
Alternative Procedure

- Be sure that **Split Plane** is **XY** and **Keep fragments** is set as **Positive Side**
- Press **OK**



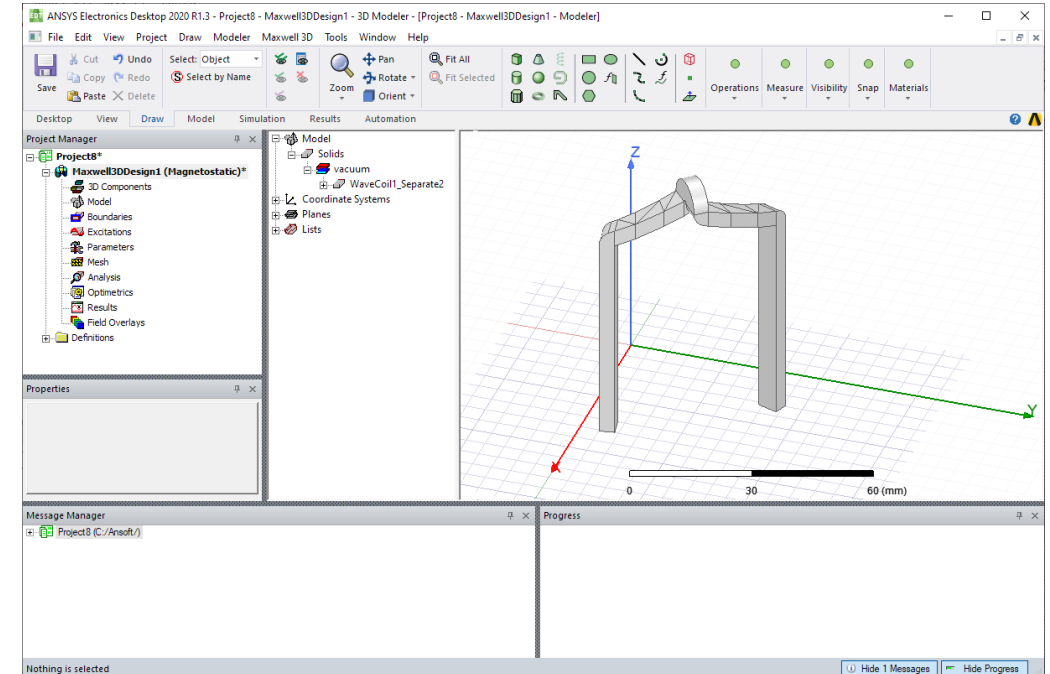
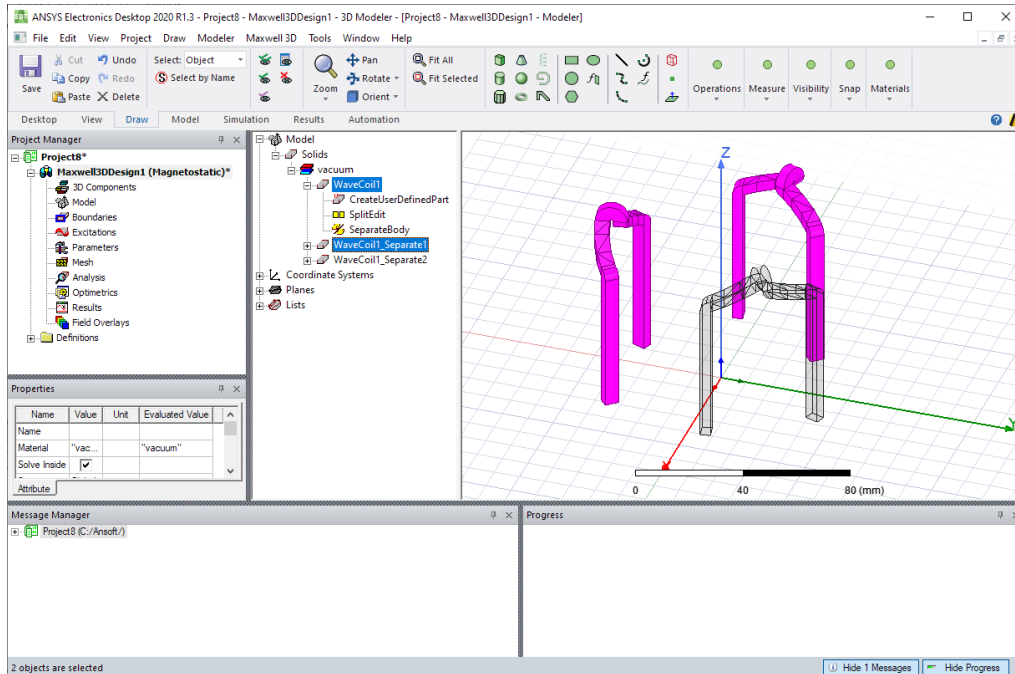
Alternative Procedure

- RMB on the *WaveCoil* → *Edit* → *Boolean* → *Separate Bodies*



Alternative Procedure

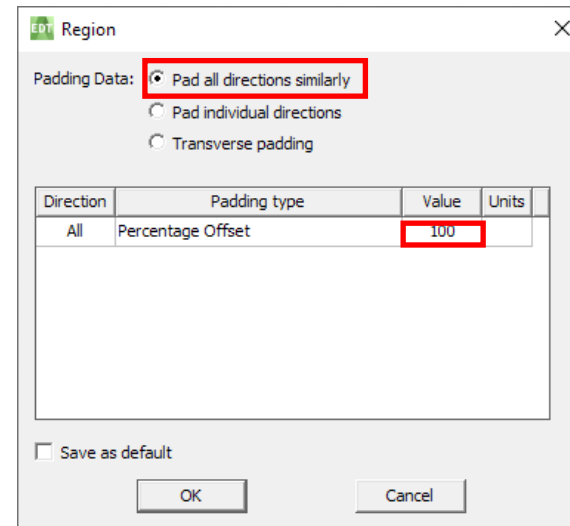
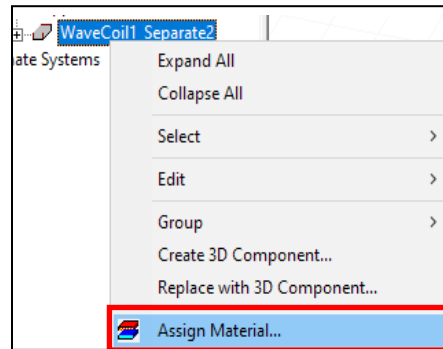
- Select the Objects **WaveCoil1** and **WaveCoil1_Separate1** (hold **CTRL** down for multiple selection)
- Delete both the selected objects



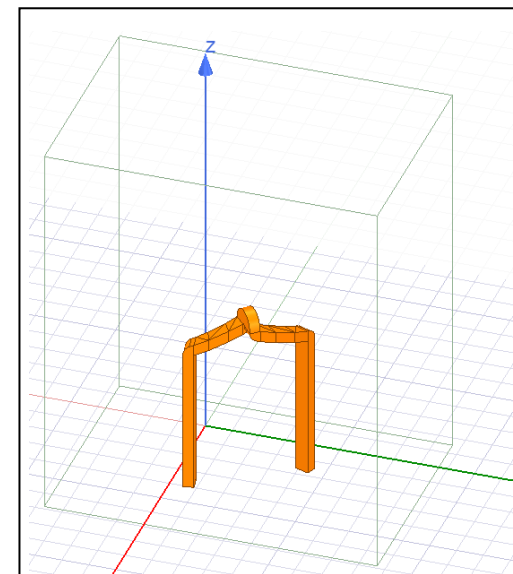
Note: you can rename the final resulting object and change its colour as you wish

Setup Problem

- **Assign Material**
 - Select the final resulting object
 - **RMB** → **Assign Material** → **Copper**
- **Create Simulation Region**
 - Select the menu item **Draw** → **Region** or click on the  icon
 - Pad all directions similarly: ☒ **Checked**
 - Padding Type: Percentage Offset
 - Value: 100
 - Press OK
 - Double click on Create Region
 - -Z Padding Data: 0



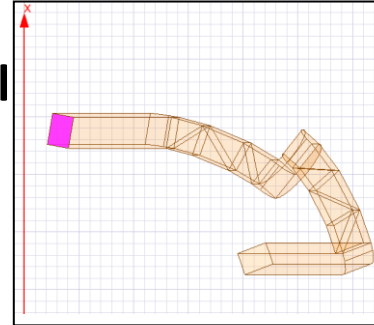
	Name	Value	Unit	Evalu...
	Command	CreateRegion		
	Coordinate Sys...	Global		
	+X Padding Ty...	Percentage Offset		
	+X Padding D...	100	100	
	-X Padding Type	Percentage Offset		
	-X Padding Data	100	100	
	+Y Padding Ty...	Percentage Offset		
	+Y Padding D...	100	100	
	-Y Padding Type	Percentage Offset		
	-Y Padding Data	100	100	
	+Z Padding Ty...	Percentage Offset		
	+Z Padding D...	100	100	
	-Z Padding Type	Percentage Offset		
	-Z Padding Data	0	0	



Setup Problem

- Assign Excitations

- Press the **F** button to select Faces
- Select one terminal section of the coil
- **RMB** → **Assign Excitation** → **Current**
 - Value: 100 A
 - Solid: ☒ Checked



Current Excitation

Name: Current1

Parameters

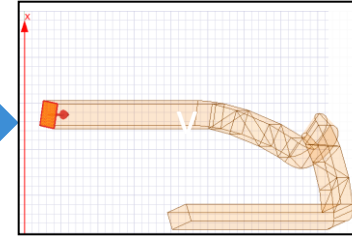
Value: 100 A

Type: ☒ Solid ☐ Stranded

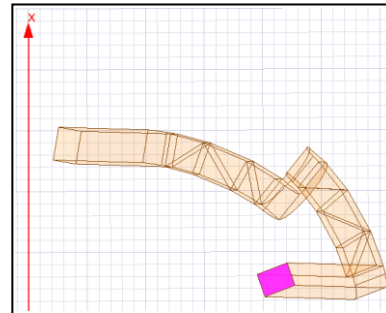
Swap Direction

Use Defaults

OK Cancel



- Select the second coil terminal section
- **RMB** → **Assign Excitation** → **Current**
 - Value: 100 A
 - Solid: ☒ Checked
 - Swap Direction



Current Excitation

Name: Current2

Parameters

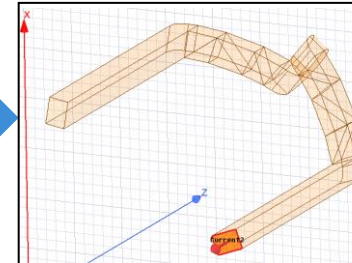
Value: 100 A

Type: ☒ Solid ☐ Stranded

Swap Direction

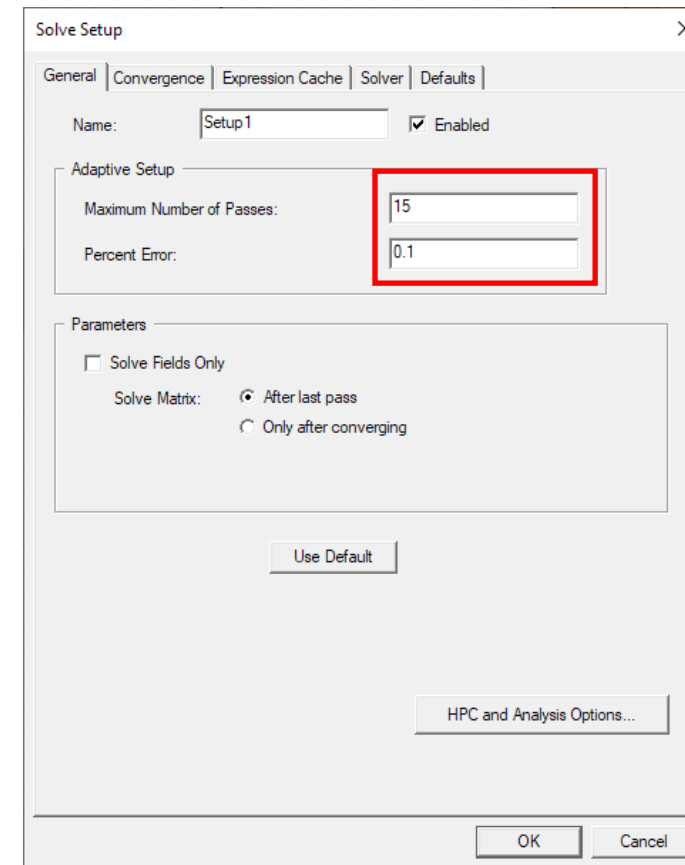
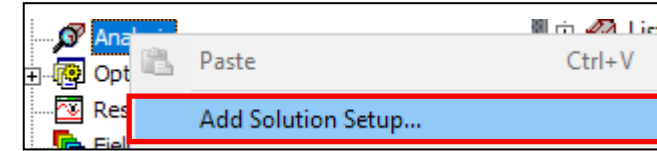
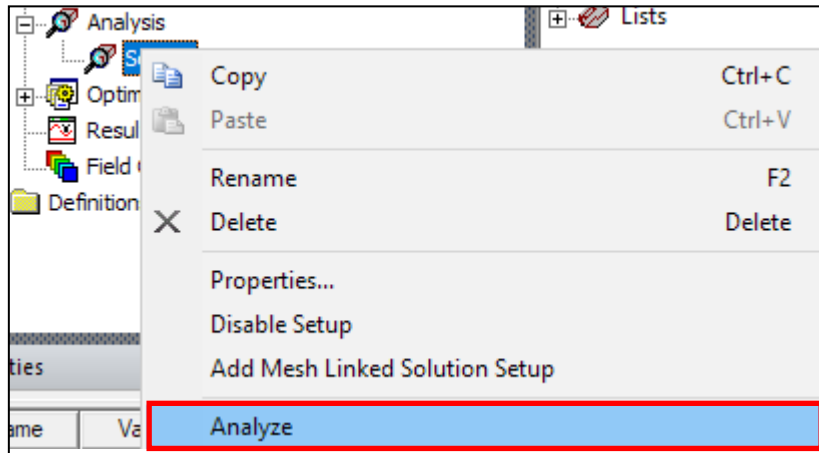
Use Defaults

OK Cancel



Analyze

- Create an analysis setup:
 - *RMB on Analysis → Add Solution Setup*
 - Solution Setup Window:
 - General Tab
 - Maximum Number of Passes: 15
 - Percent Error: 0.1%
 - Click the OK button
- Start the solution process:
 - *RMB on Setup1 → Analyze*

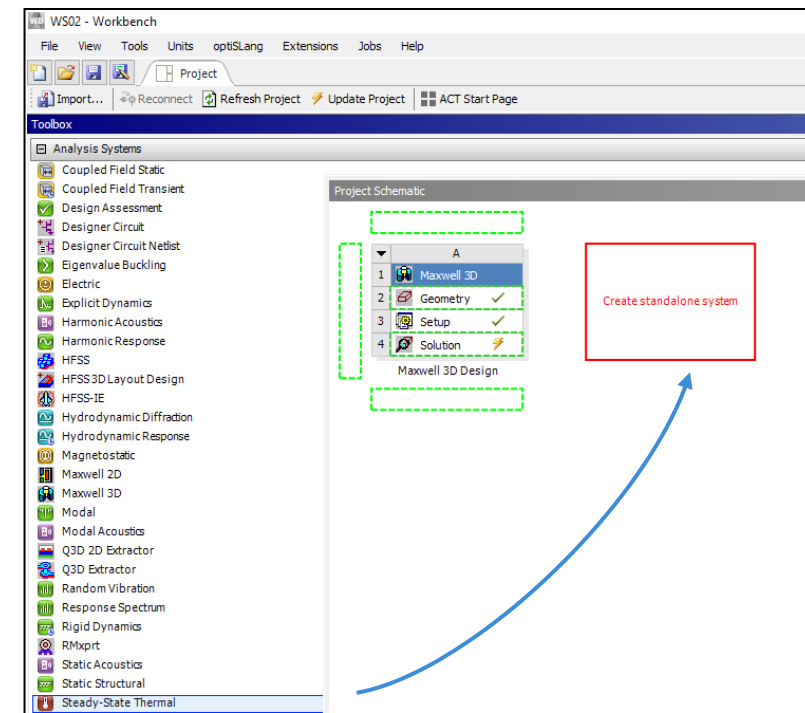


Insert Steady State Thermal Analysis

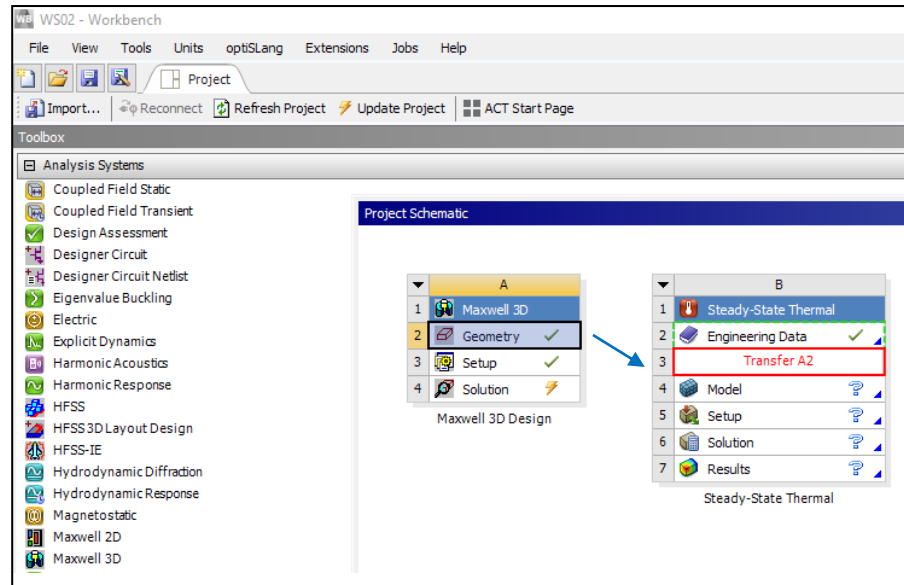
• Select Steady State Thermal

- drag and drop a Steady-State Thermal analysis system into the Project Schematic
- Select the Maxwell Geometry cell **A2** and drag it on the Steady-State Thermal Geometry cell **B3**
- Repeat the material assignment procedure explained in Workshop 01

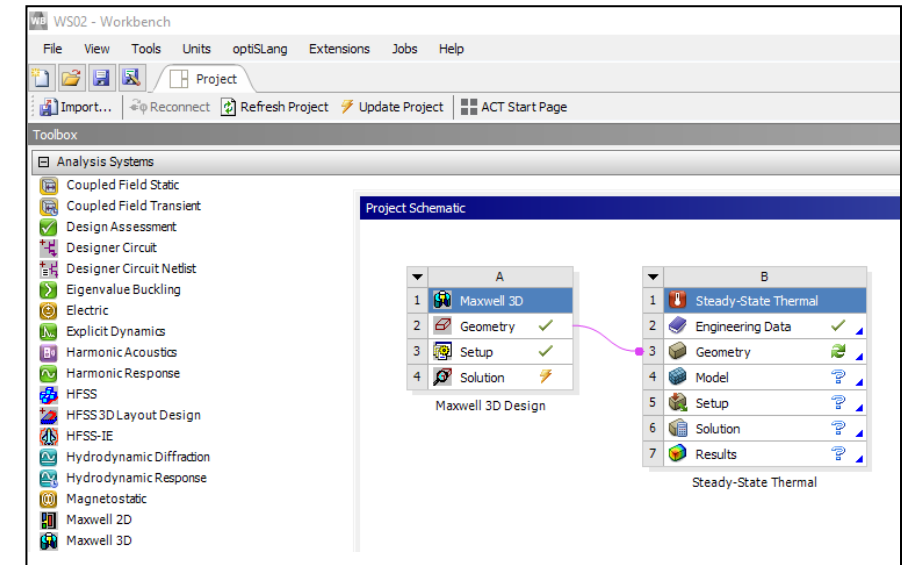
1



2



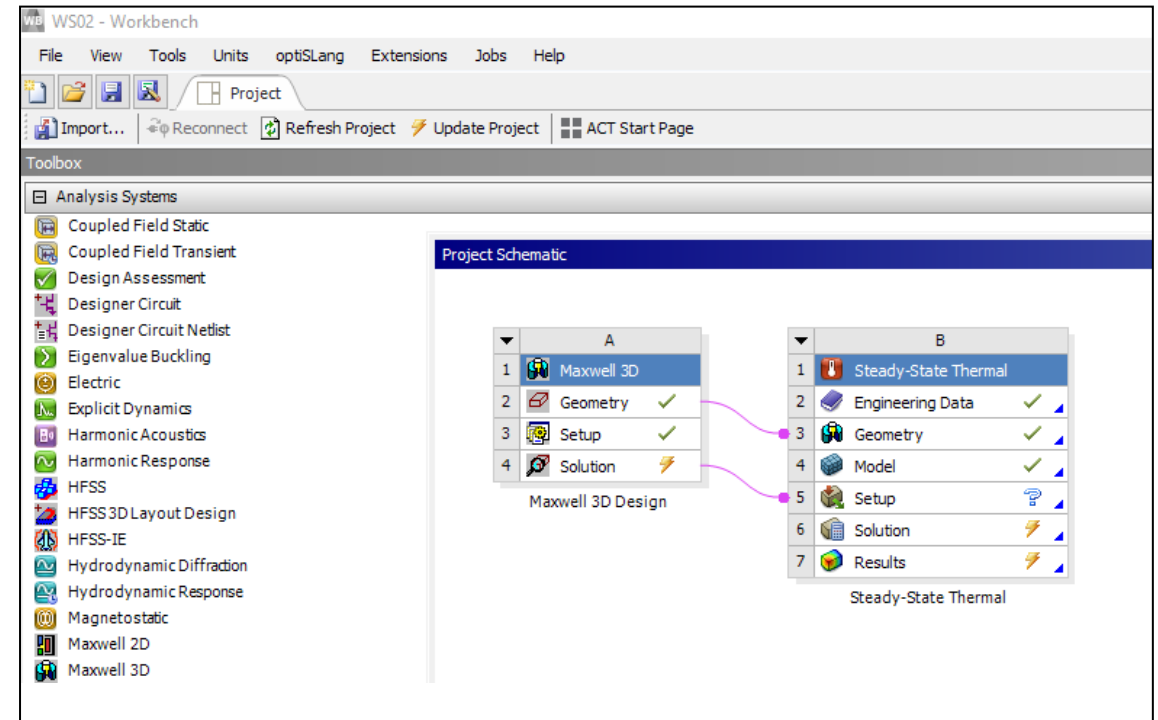
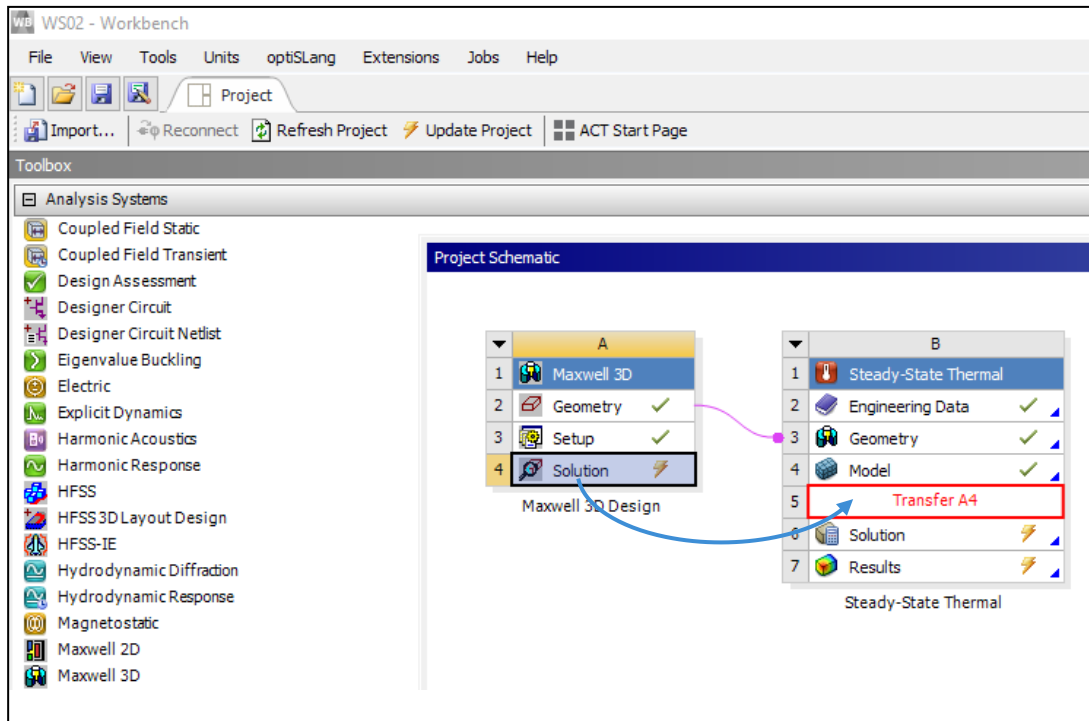
3



Connect Maxwell Solution and Steady State Thermal Analysis Setup

- Select Maxwell

- drag and drop Maxwell solution cell **A4** on cell **B5** (Setup)
- Update cell **A4** (Maxwell Solution) and cell **B4**
- Double click on cell **B5** to open Ansys Steady State Thermal

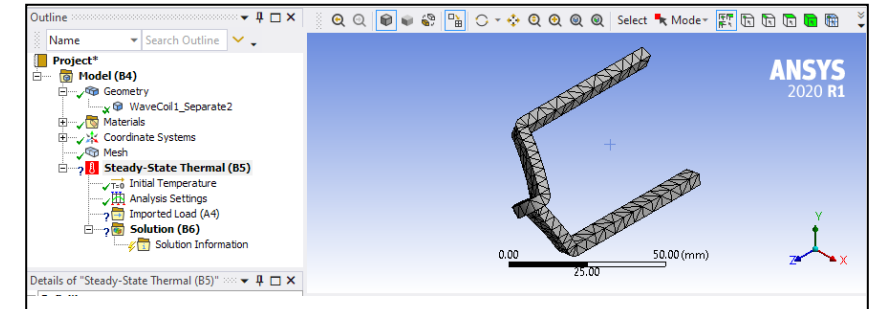


Import the load from Maxwell

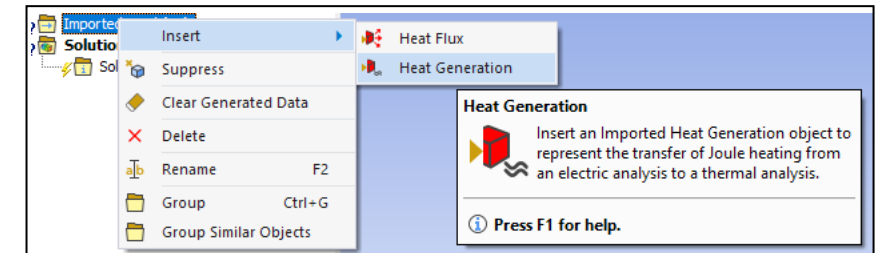
- In Ansys Steady State Thermal

- Click F7 or icon  to fit the view
- *RMB on Imported Load (A4) → Insert → Heat Generation*
- Select the geometric body
- Under Details select Geometry and Apply
- *RMB on Imported Heat Generation → Import Load*

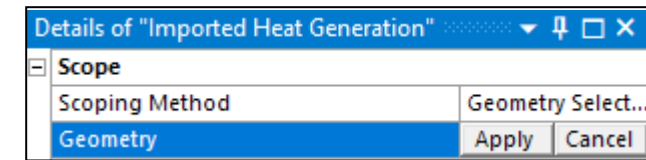
1



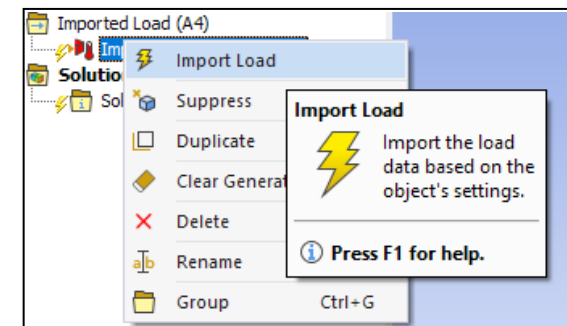
2



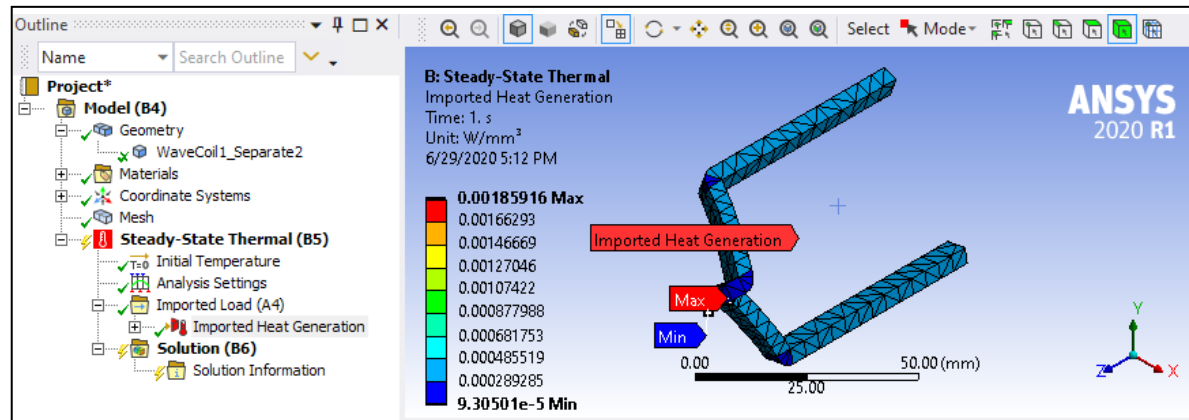
3



4

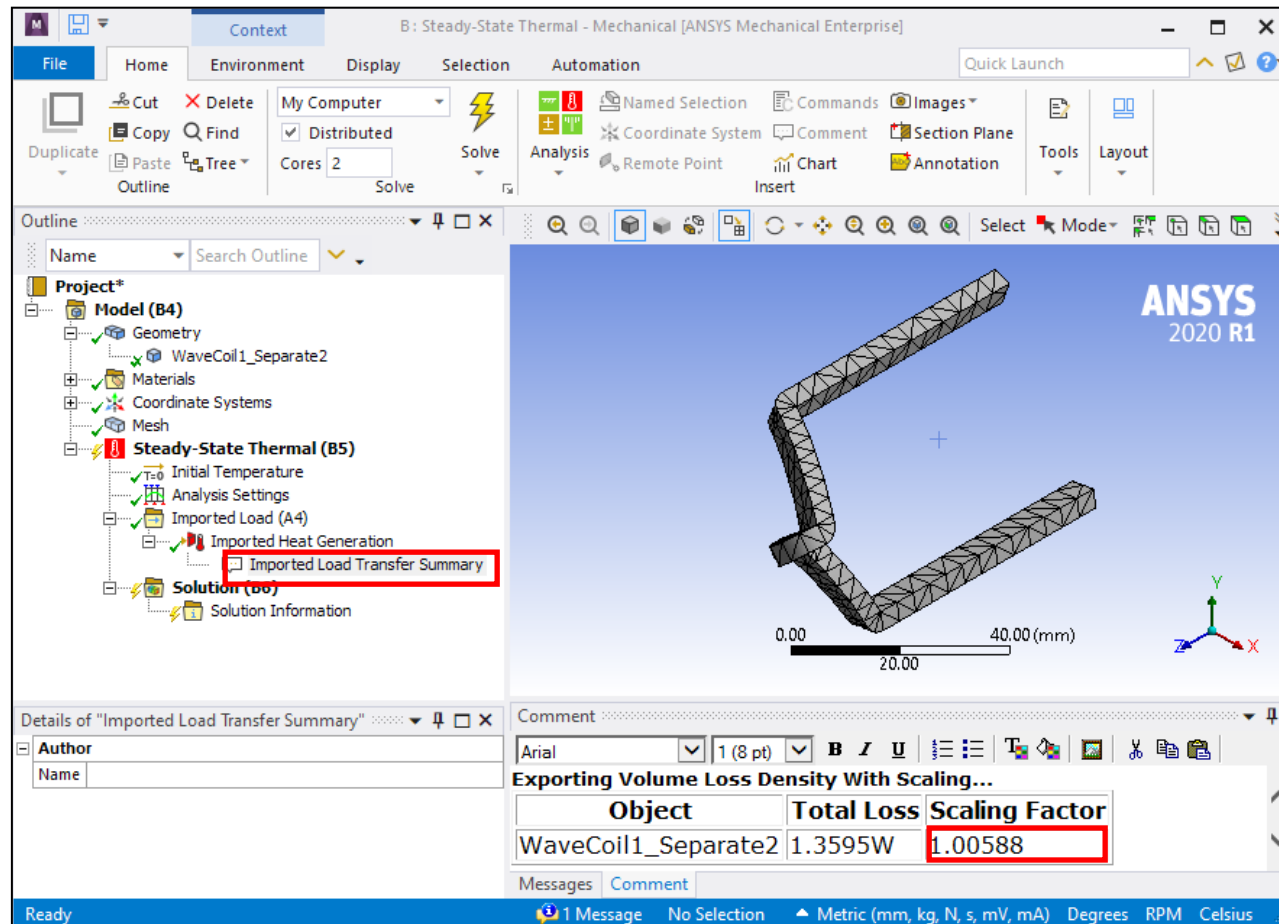


5



Inspect Imported Load From Maxwell

- Under Imported Heat Generation select **Imported Load Transfer Summary**
- Be sure the **Scaling Factor** is near to **1**. Values much different from 1 indicate a bad data transfer from Maxwell to Ansys Structural, most often due to a bad correspondence between the two meshes

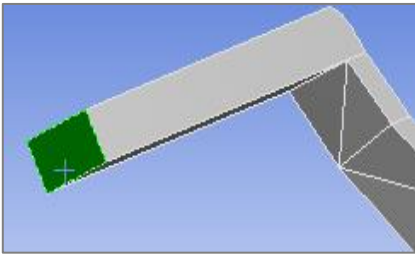


/ Set Thermal Boundaries

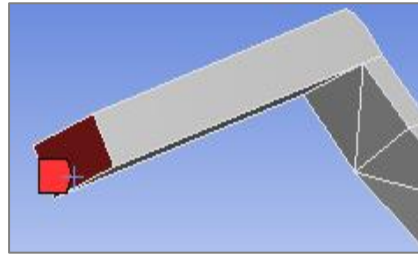
- In Ansys Steady State Thermal

- Click the icon  to select the faces
- Select one coil terminal cross section
- *RMB on Steady State Thermal (B5) → Insert → Temperature*
- Select the other coil terminal cross section
- *RMB on Steady State Thermal (B5) → Insert → Temperature*

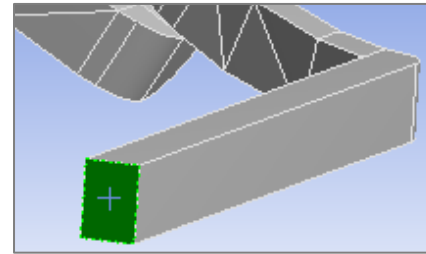
1



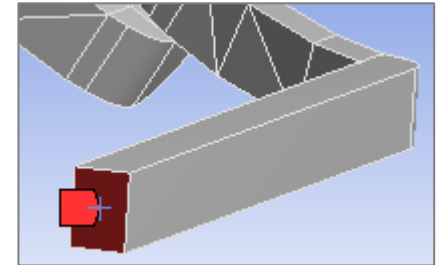
2



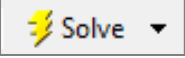
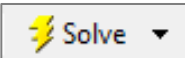
3

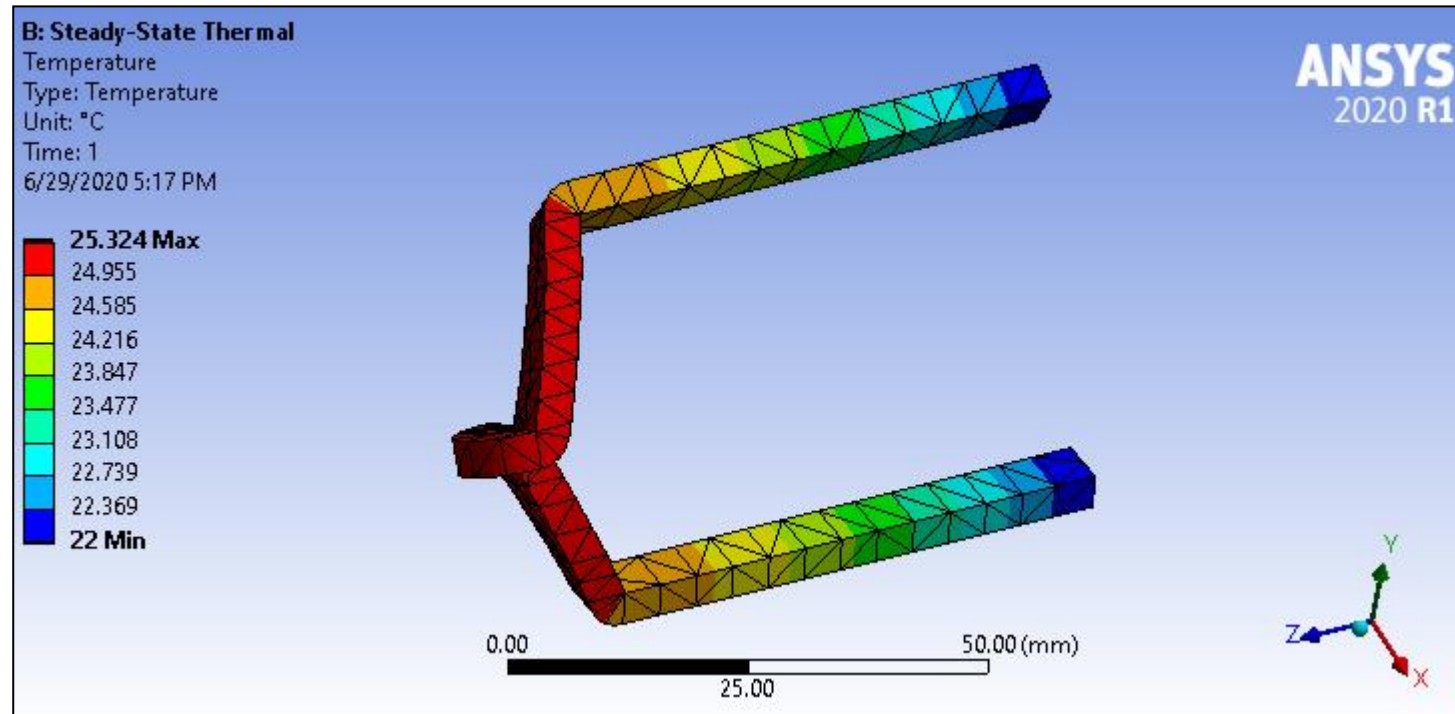


4



/ Solve the Thermal Design

- In Ansys Steady State Thermal
 - Click the icon  to solve the Thermal Analysis
 - *RMB on Solutions (B6) → Insert → Thermal → Temperature*
 - Click the icon  to evaluate results
 - Temperature distribution is then shown graphically



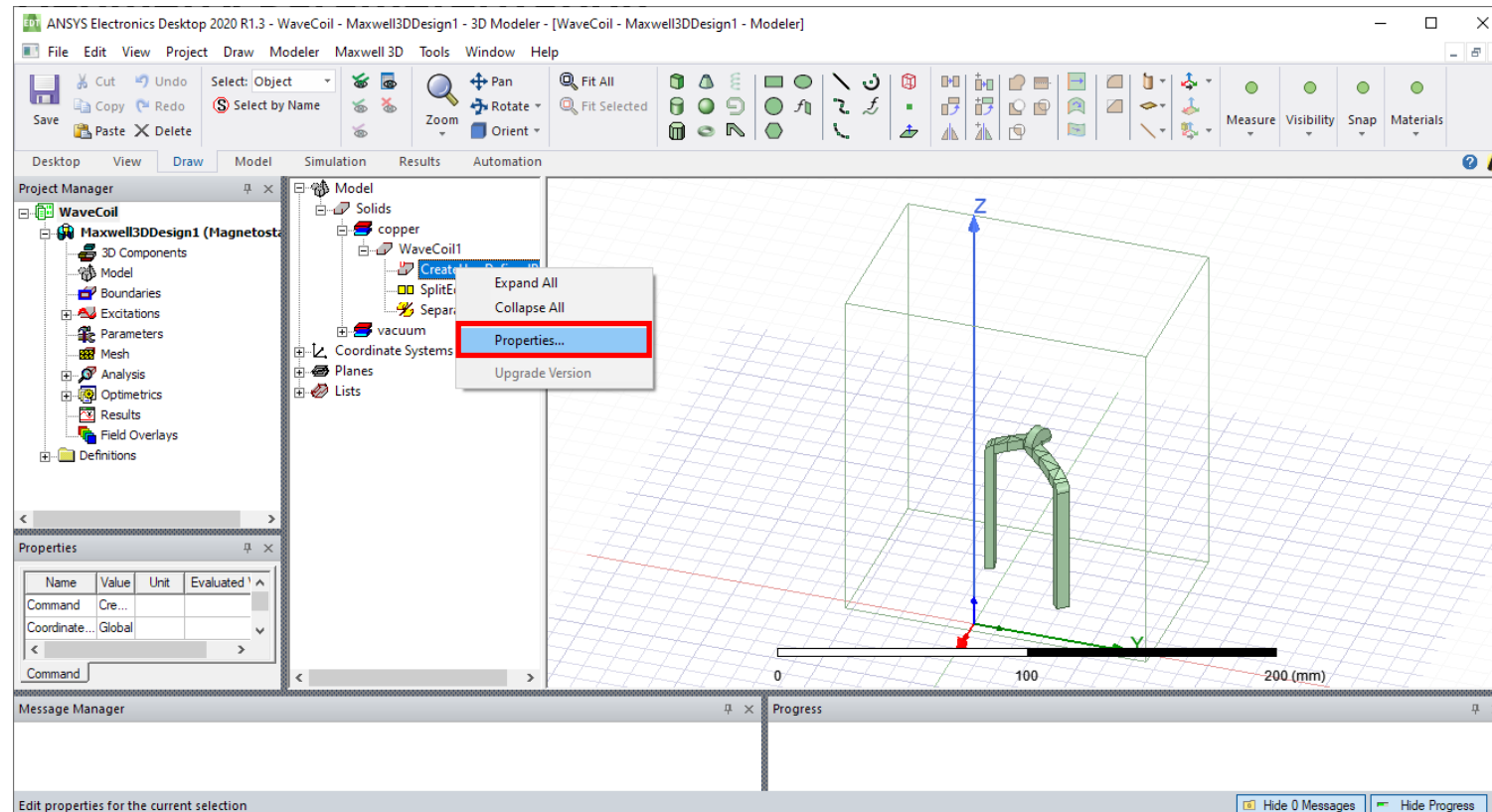
Saving the Project

- This completes the workshop
- Save the file with the name **Workshop 05.2** in the working folder

Appendix: Homework – Parameter set

Geometry Parameterization

- In this section we will parameterize the initial geometry and will share the parameters with the workbench, so that it will be possible to study the influence of those parameters on thermal results
- To do that, please create the geometry as explained in slides from 6 to 10 and implement all the settings as for slides 11-13
- Select the item **CreateUserDefinedPart** from the History tree and **RMB** → **Properties**



Geometry Parameterization

- In the UDP Properties window, type “**Length**” in the field *Length* and press the *Enter* key
- In the **Add Variable** window, press **OK**

Properties: WaveCoil - Maxwell3DDesign1 - Modeler

Command

Name	Value	Unit	Evaluated Value	Description
Command	CreateUserDefinedPart			
Coordinate Sys...	Global			
Name	RMxprt/WaveCoil.dll			
Location	syslib			
Version	12.1			
DiaGap	100	mm	100mm	Core diameter on gap si...
DiaYoke	20	mm	20mm	Core diameter on yoke ...
Length	Length	mm	100mm	Core length
Skew	0	deg	0deg	Skew angle in core len...
Slots	18		18	Number of slots
Slot Type	1		1	Slot type: 1 to 7
Hs0	1	mm	1mm	Slot opening height
Hs1	1	mm	1mm	Slot wedge height
Hs2	10	mm	10mm	Slot body height
Bs0	2.5	mm	2.5mm	Slot opening width
Bs1	8	mm	8mm	Slot wedge maximum w...
Bs2	5	mm	5mm	Slot body bottom width...
Rs	0	mm	0mm	Slot body bottom fillet
FilletType	0		0	0: a quarter circle; 1: ta...
Layers	2		2	Number of winding layers
CoilPitch	3		3	Coil pitch measured in s...
EndExt	5	mm	5mm	One-side end extended...
SpanExt	18	mm	18mm	Axial length of end spa...
SegAngle	10	deg	10deg	Deviation angle for end...
PolePitch	6		6	Pole pair pitch measure...
LenRegion	200	mm	200mm	Region length
InfoCoil	1		1	0: winding; 1: one coil; ...

☐ Show Hidden

OK Cancel Apply



Add Variable

Name

Unit Type

Unit

Value

Type

OK Cancel

Geometry Parameterization

- In the UDP Properties window, type “**Dim_1**” in the field **Hs2** and press the **Enter** key
- In the **Add Variable** window, press **OK**
- Press **OK** to close the UDP Properties window

Properties: WaveCoil - Maxwell3DDesign1 - Modeler

Command

Name	Value	Unit	Evaluated Value	Description
Command	CreateUserDefinedPart			
Coordinate Sys...	Global			
Name	RMxpri/WaveCoil.dll			
Location	syslib			
Version	12.1			
DiaGap	100	mm	100mm	Core diameter on gap si...
DiaYoke	20	mm	20mm	Core diameter on yoke ...
Length	Length		100mm	Core length
Skew	0	deg	0deg	Skew angle in core len...
Slots	18		18	Number of slots
SlotType	1		1	Slot type: 1 to 7
Hs0	1	mm	1mm	Slot opening height
Hs1	1	mm	1mm	Slot wedge height
Hs2	Dim_1	mm	10mm	Slot body height
Bs0	2.5	mm	2.5mm	Slot opening width
Bs1	8	mm	8mm	Slot wedge maximum w...
Bs2	5	mm	5mm	Slot body bottom width,...
Rs	0	mm	0mm	Slot body bottom fillet
FilletType	0		0	0: a quarter circle; 1: ta...
Layers	2		2	Number of winding layers
CoilPitch	3		3	Coil pitch measured in s...
EndExt	5	mm	5mm	One-side end extended...
SpanExt	18	mm	18mm	Axial length of end spa...
SegAngle	10	deg	10deg	Deviation angle for end...
PolePitch	6		6	Pole pair pitch measure...
LenRegion	200	mm	200mm	Region length
InfoCoil	1		1	0: winding; 1: one coil; ...

☐ Show Hidden

OK Cancel Apply



Add Variable

Name: Dim_1

Unit Type: Length

Unit: mm

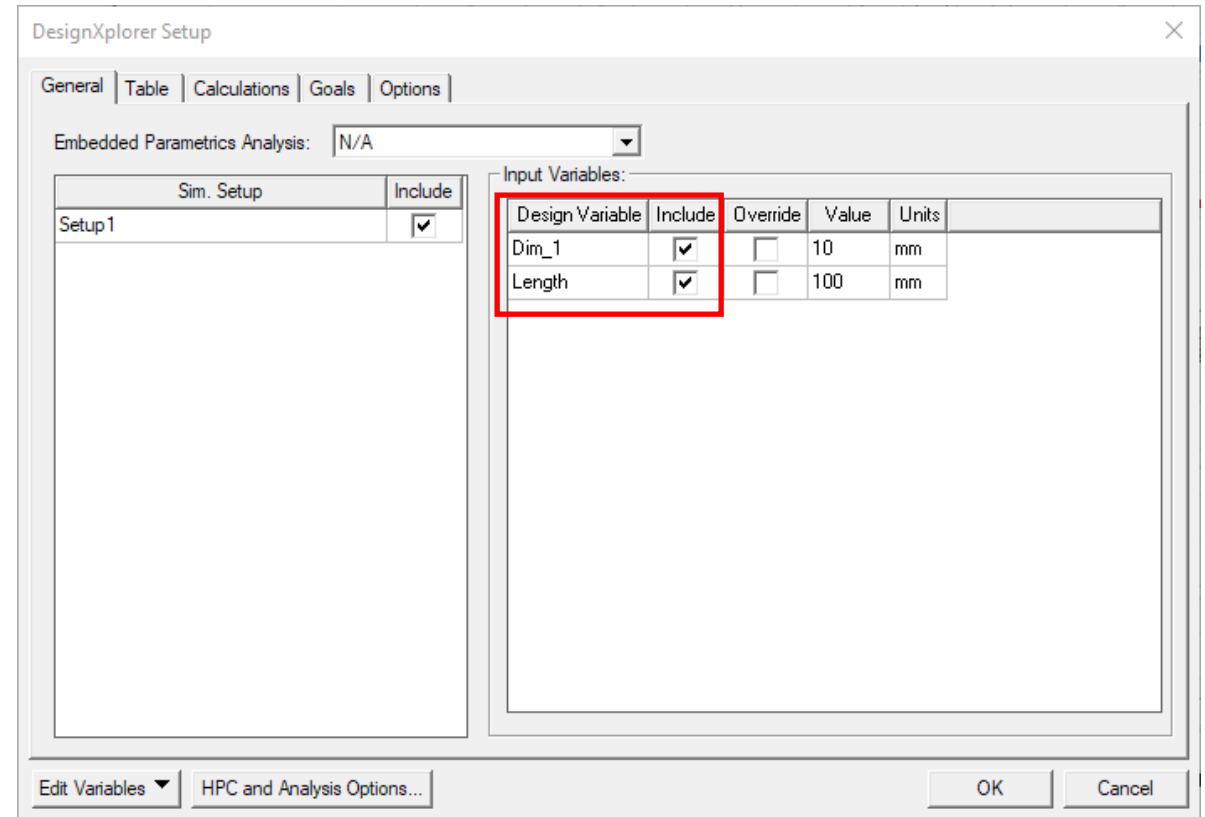
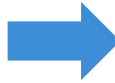
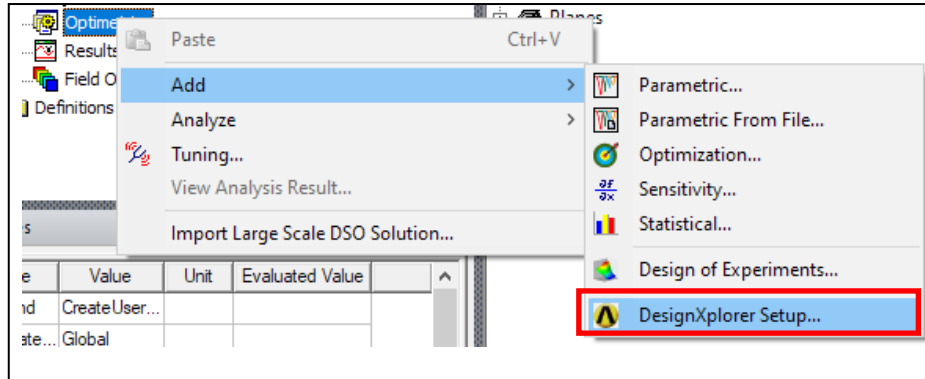
Value: 10

Type: Local Variable

OK Cancel

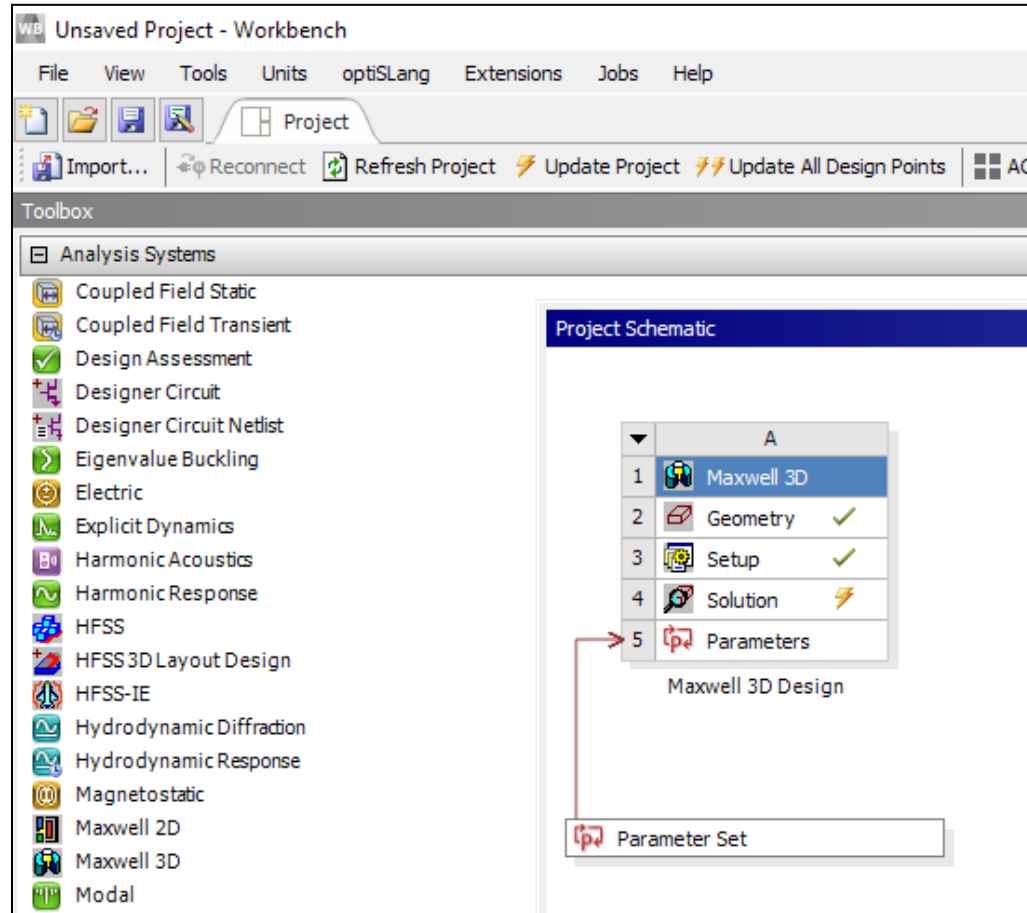
Geometry Parameterization

- In the Project Manager window, RMB on **Optimetrics** and click on **DefaultDesignXplorerSetup**
- In the **DesignXplorer Setup** window, check Include for both the **Dim_1** and **Length** variables
- Press **OK** to close the **DesignXplorer Setup** window



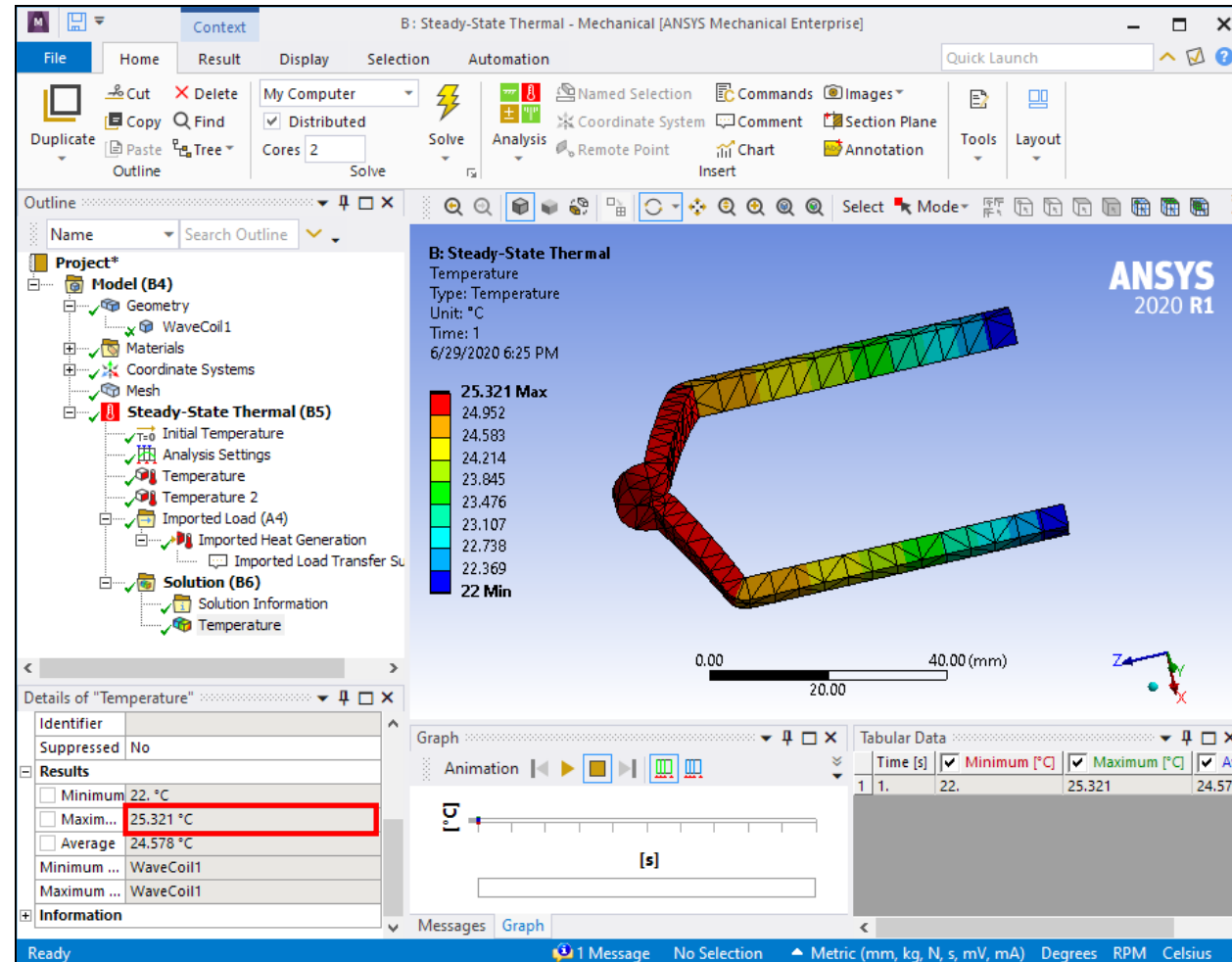
Parameter Set

- A new object named Parameter Set appear in the workbench interface
- The two variables are now present in the workbench as global variables and can be used through the different physical domains



Solve the Project

- Follow the instructions explained in slides 14-18, inserting the Thermal Steady State System, importing the load from Maxwell and solving the thermal problem
- As already shown, the final result is the following, where the maximum temperature is about 25.3 °C



Change the Parameters

- Double click on the **Parameter Set** to access the variables editing
- Change **P2 – Dim_1** to 5 mm and then close the **Parameter Set** Tab

The screenshot shows the 'Parameter Set' dialog box in Ansys Workbench. The 'Outline of All Parameters' table is as follows:

	A	B	C	D
1	ID	Parameter Name	Value	Unit
2	Input Parameters			
3	Maxwell 3D Design (A1)			
4	P1	Length [mm]	100	
5	P2	Dim_1 [mm]	5	
*	New input parameter New name New expression			
7	Output Parameters			
*	New output parameter New expression			
9	Charts			

The 'Table of Design Points' is as follows:

	A	B	C	D	E
1	Name	P1 - Length [mm]	P2 - Dim_1 [mm]	Retain	Retained Data
2	DP 0 (Current)	100	5	☒	☒
*				☐	

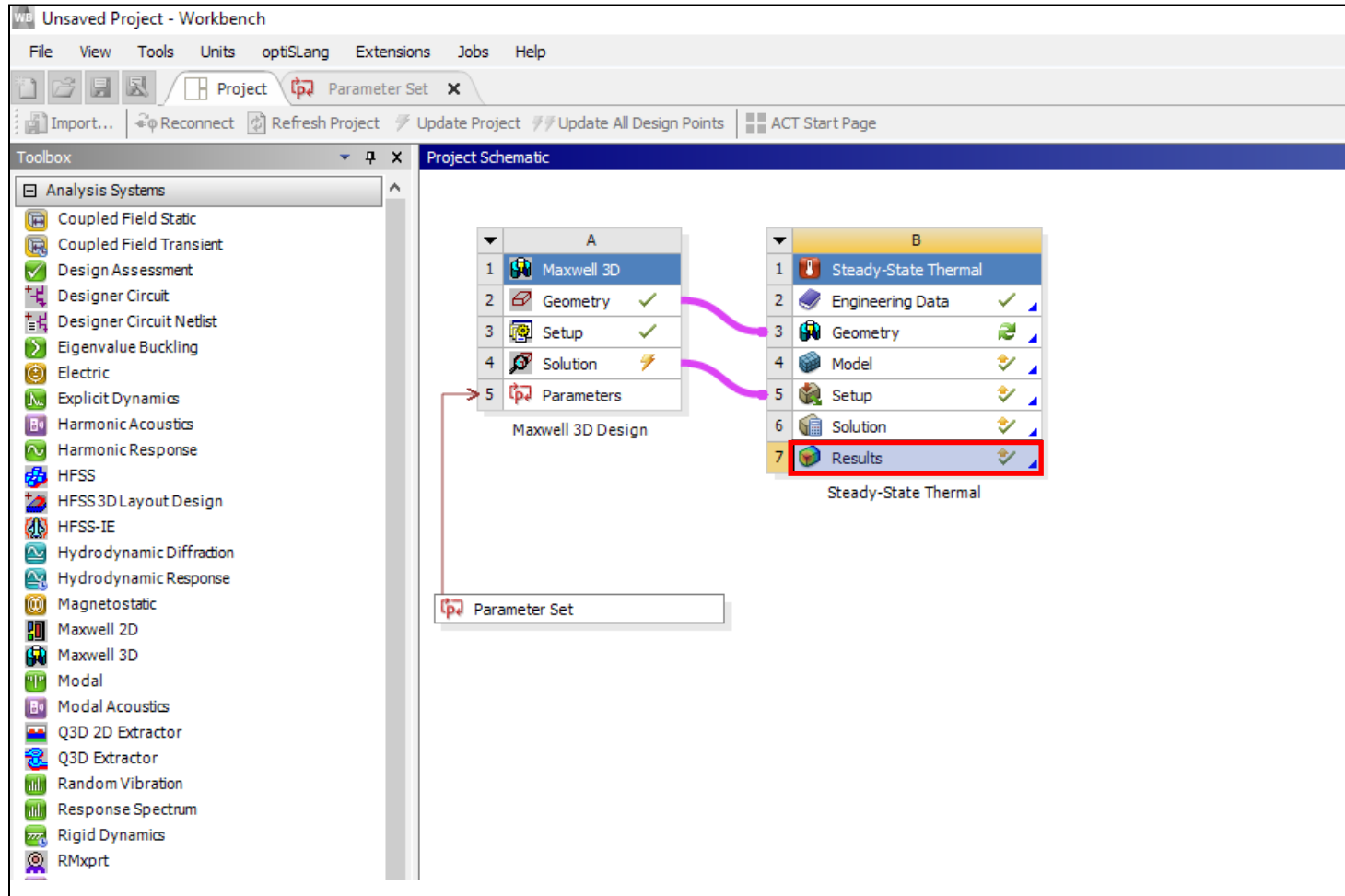
The 'Properties of Outline C5: P2' table is as follows:

	A	B
1	Property	Value
2	General	
3	Expression	5
4	Usage	Input
5	Description	
6	Error Message	
7	Expression Type	Constant
8	Quantity Name	

Note: decreasing Dim_1 value has the effect to decrease the conductor section and thus increase the conductor resistance

Update the Project

- Go to Cell **B7 – Results** of the Steady State Thermal System, **RMB** and select **Update**



Inspect the Results

- The results have been automatically updated: double click on Cell **B7 – Results** to inspect the new temperature distribution
- Now the maximal temperature is around **30.3 °C**, as expected since the conductor section has been decreased (see slides 27)

